Experiment 10

Performance Optimization of the Student Admission Application

Aim

To optimize the performance of the Student Admission Enquiry Management (Advanced Edition) application in Zoho Creator by improving form loading, reducing unnecessary database operations, optimizing Deluge scripts and ensuring efficient report generation.

Procedure

1. Log in to Zoho Creator and open the Student Admission Enquiry Management (Advanced Edition) application.

2. Review the Admission Enquiry Form and remove unnecessary fields, scripts and workflows.

3. Optimize Deluge scripts by avoiding repeated database queries and unnecessary loops.

4. Use criteria-based record fetching to retrieve only required enquiries.

5. Optimize reports by displaying only essential columns and applying filters.

6. Reduce unnecessary workflow executions and duplicate notifications.

7. Test the application after changes and compare loading speed and responsiveness.

8. Verify that all admission-enquiry features work correctly after optimization.

Optimization Actions

Area	Before	After
Form	All advanced fields loaded without grouping	Fields grouped into logical sections E–H
Lookup	Repeated school lookups	Criteria-based lookup to required schools
Reports	All columns displayed	Essential columns plus filters
Workflows	Runs on every edit	Runs only when relevant admission fields change

Deluge Optimization Example

BEFORE:

```
for each r in Admission_Enquiry_Form
{
  if(r.Scholarship_Eligibility_Percent > 0)
  {
    // process every record unnecessarily
  }
}
```

AFTER:

```
eligible = Admission_Enquiry_Form[Scholarship_Eligibility_Percent > 0];
for each r in eligible
{
  // process only matching enquiries
}
```

Optimization Flow

Figure 10.1: Optimization approach for the Admission Enquiry application

Before and After Comparison

Area	Before	After	Benefit
Form	Unstructured advanced fields	Grouped sections	Faster and clearer data entry
Queries	All enquiries fetched	Criteria-based fetch	Less database work
Reports	Many columns, no filters	Essential columns and filters	Quicker report display

Area	Before	After	Benefit
Workflows	Trigger on every edit	Trigger on relevant changes	Fewer duplicate actions

Verification Checklist After Optimization

Check	Expected Result
Submit admission enquiry	Record saved successfully
Change scholarship percentage	Net fee recalculates correctly
Open Admission_Enquiry_Report	Essential columns and filters load
Open School Name Lookup	Relevant schools appear quickly

Result

The Student Admission application was optimized by grouping advanced fields, using criteria-based fetching, trimming report columns and reducing unnecessary workflow executions.